

Christian Diaz

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Education

University of Massachusetts Amherst | Commonwealth Honors College

Amherst, MA

Bachelor of Science in Computer Science

August 2024 – May 2027

- **Relevant Coursework:** Artificial Intelligence; Web Programming; Data Structures; Probability & Statistics; C Programming; Data Management; Machine Learning

Professional & Technical Experience

Cytrence

Burlington, MA

Embedded Systems Intern

Jun 2026 – Aug 2026

- Developed Python-based automation for connected hardware systems, integrating device detection, multithreading, logging, screenshot capture, and keyboard/mouse control across multiple computing environments.
- Built OpenCV and NumPy computer-vision tooling to analyze screenshots, distinguish meaningful interface changes from visual noise, and validate input behavior across operating-system and firmware environments.
- Prototyped automation for interacting with firmware interfaces, combining computer vision and Python-based control logic to handle varying system interfaces.
- Created a short-form technical product demo showcasing Kiwi connectivity across computing ecosystems and demonstrating core product capabilities, reaching 220K+ combined views across YouTube and Instagram.

Pelican Point East

Naples, FL

Freelance Full-Stack Developer

2024 – Present

- Designed and developed a production full-stack product for a condominium association, translating resident and board requirements into document-access, administrative, and publishing workflows reaching 1,000+ users.
- Built a Node.js/Express backend with PostgreSQL and Firebase, implementing bcrypt password hashing, JWT authentication, role-based authorization, and manager-controlled user administration.
- Developed resident and administrative workflows for documents, financial reports, and notices, including publishing, filtering, metadata management, and access controls supporting Florida document-access requirements.
- Improved platform discoverability through structured metadata and SEO optimization; 43% of users were acquired through organic search.

Learning Resource Center

Amherst, MA

Supplemental Instruction Leader

Aug 2025 – Jan 2026

- Led two weekly Supplemental Instruction sessions for CICS 110, developing review materials, debugging exercises, and guided practice for a course serving 200+ students per semester.
- Collaborated with course staff to identify recurring student difficulties and adapted explanations and exercises to strengthen programming fundamentals and problem-solving skills.

Projects & Leadership

BUILD UMass

Amherst, MA

Website Project Lead / Software Developer

Sep 2024 – Sep 2026

- Led modernization of BUILD UMass's website, migrating the frontend from JavaScript/React-Bootstrap to TypeScript and Tailwind CSS while rebuilding core pages and reusable components in Next.js.
- Led development of an end-to-end recruiting application system, replacing external forms with custom Software Developer and Product Manager workflows backed by Prisma/PostgreSQL and a protected admin dashboard, supporting 400+ applications in its first year.
- Contributed to a full-stack retrieval-augmented generation (RAG) prototype for the New England First Amendment Coalition, improving React/TypeScript handling of streamed AI responses and result state for First Amendment resource discovery.

Maintenance Report Board

Full-Stack Web Application

Jul 2026 – Aug 2026

- Built a maintenance-report backend in Node.js/Express with server-rendered EJS views and a controller-service-repository architecture for validated report submission and management.
- Implemented session-based authentication and role/ownership authorization, and developed HTMX partial-page interactions with a responsive Tailwind CSS interface.

Technical Skills

Languages: Python; TypeScript; JavaScript; SQL; Java; C; HTML/CSS

Frameworks & Development: React; Next.js; Node.js; Express; FastAPI; Jest; Tailwind CSS; Git; Docker

Data, AI & Platforms: PostgreSQL; MongoDB; Prisma; Firebase; OpenCV; NumPy; Pandas; Hugging Face; Linux; AWS

Languages Spoken: English and Spanish (native fluency)